

The inversion law — [*FROZEN* INTERNAL NOTE, NOT *PUBLICATION – READY*]

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Abstract

Dehn surgery on a knot complement produces a closed 3-manifold — the *child* — and forgets its *parent*: distinct knots can be surgered to the same child. We study the child of the figure-eight knot at slope (5,1) and prove, over a year of invariants, that it cannot name its parent: its geometry, its trace field, its Chern–Simons value, its quantum invariants are all class-generic — they see the child’s own arithmetic, not the knot it came from. The figure-eight’s invariant trace field $\mathbb{Q}(\sqrt{-3})$ is *absent* from the child: the child’s trace field is the quartic $\mathbb{Q}[x]/(x^4 - x - 1)$, a field with Galois group S_4 and **no quadratic subfield at all**, so no quadratic parent-field can survive. What can survive a collision is at most a single bit — the **orientation** — and even that survives only sometimes. The collision $4_1(5,1) \cong 5_2(5,1)$ is orientation-reversing: the two children are a mirror pair (conjugate geometric embeddings of the same quartic, Chern–Simons ± 0.07703818), so here orientation is the one datum that distinguishes the parents. But the twist-knot *triple* $4_1(1,2) = 5_2(-1,1) = 6_1(1,1)$ is orientation-preserving (one oriented manifold, equal Chern–Simons), so there orientation distinguishes nothing. The surviving parental data is therefore zero or one bit, never more, and which of the two depends on the collision. That is the inversion law: surgery on this family launders the parent’s arithmetic completely, and what is left is at most orientation.

FROZEN (2026-07-09) as an internal note. After three adversarial re-panel rounds, this write-up still carries *framing/labelling* defects (documented in `SCRUTINY_P1P3_round3.md`) — notably that the “ $\leq$ one bit” law is partly tautological (a collision is *defined* by a shared unoriented child). Retained as an internal record only; **the verified computations live in the cited B-nodes** (B435–B443, B467). Do not circulate as a paper.

1 Introduction

If you glue a solid torus into a knot complement, you get a closed manifold, and you throw a question away: *which knot?* Two different knots, surgered along the right slopes, land on the same closed manifold, and the manifold keeps no record of the difference. This paper is a year of asking one such manifold — the child of the figure-eight at slope (5,1) — to name its parent, and the accumulating proof that it cannot, together with an honest account of the single bit that sometimes, but not always, survives.

The temptation, at the start, was that the figure-eight is special — arithmetic, the simplest hyperbolic knot, its trace field the Eisenstein field $\mathbb{Q}(\sqrt{-3})$ — and that its specialness would survive into the child and mark it. It does not. Every invariant we computed reads the child’s own class data, not the parent’s memory. The field $\mathbb{Q}(\sqrt{-3})$ that defines the figure-eight does not appear in the child at all; indeed the child’s trace field has no quadratic subfield to hold it. Surgery is a forgetting, and it forgets thoroughly. The one candidate for a survivor is

orientation — and here the story is subtler than we first wrote it. Sometimes the collision that merges two knots into one child is a mirror, and then orientation is exactly the surviving distinction. Sometimes the collision is orientation-preserving, and then even orientation is gone. Section 2 sets up the child and its invariants; Section 3 is the laundering — the systematic absence of parental data; Section 4 is the census and the *non-uniform* orientation signature; Section 5 is the law, stated for what it forbids and honest about what it keeps.

2 The child and its invariants

The figure-eight complement 4_1 has one cusp; the surgery slope (p,q) fills it, producing the closed manifold $4_1(p,q)$. At slope $(5,1)$ the child is hyperbolic, with volume 0.9813688, $H_1 = \mathbb{Z}/5$, and a well-defined geometry. Its measurable invariants: hyperbolic volume; the trace field of its holonomy; the Chern–Simons invariant $CS \in \mathbb{R}/\mathbb{Z}$; and the Reshetikhin–Turaev /Witten quantum invariants at roots of unity.

The parents are knots K with a slope σ_K such that $K(\sigma_K) \cong 4_1(5,1)$. The census supplies them: the figure-eight at $(5,1)$ and the knot 5_2 at $(5,1)$ are the same closed manifold, both equal to the census manifold $m003(-2,3)$.

For orientation bookkeeping, recall the diagnostic used throughout: for an oriented hyperbolic 3-manifold, the trace field carries a distinguished *geometric* complex embedding (the one through which the holonomy is read), and the mirror image carries its complex conjugate. Two children with the same abstract trace field but conjugate geometric embeddings are mirror images; two with the same embedding are the same oriented manifold. Chern–Simons is the numerical shadow of the same bit: $CS(M) = -CS(M)$.

3 Laundering: the parent is not in the child

Theorem 3.1 (class-generic geometry; the field is laundered). *The child’s trace field is an invariant of the child’s own commensurability class, not of the parent. Concretely, the trace field of $4_1(5,1)$ is the quartic field $\mathbb{Q}[x]/(x^4 - x - 1)$, whose Galois group is S_4 (order 24, CC-verified; discriminant -283). Because the point-stabilizer S_3 is maximal in S_4 , this quartic field has **no proper subfield at all** — in particular no quadratic subfield. So the child’s field is an S_4 – generic quartic, sharing no subfield with the parent’s $\mathbb{Q}(\sqrt{-3})$: the parent’s arithmetic is laundered away by the filling.*

Remark. Remark (avoiding a vacuity trap). One must not confuse two fields of the figure-eight. Its invariant trace field is $\mathbb{Q}(\sqrt{-3})$ (the Eisenstein field, disc -3 ; 4_1 is the arithmetic manifold of $PSL_2(O_3)$). The golden field $\mathbb{Q}(\sqrt{5})$ that appears everywhere else in this program is the field of the monodromy eigenvalue — the metallic mean $\lambda_1 = \varphi$ — a different object living upstairs in the mapping-class data, not in the hyperbolic trace field. Now, because the child has no quadratic subfield, the bare statement “field X is absent from the child” is vacuous for every quadratic X — including $\mathbb{Q}(\sqrt{-3})$ — so the laundering content is not “a particular field is missing” (an earlier draft’s $\mathbb{Q}(\sqrt{5}) \rightarrow \mathbb{Q}(\sqrt{-3})$ fix merely traded one vacuous absence for another). The honest, non-vacuous statement is structural: the child’s trace field is an S_4 – generic quartic with no subfield shared with the parent — the parent’s field is not recoverable because the child’s field has no room* to contain it, and the two share nothing below the full quartic.

Theorem 3.2 (the vacua and their Chern–Simons). *The child carries four flat $SL(2, \mathbb{C})$ vacua; the geometric one reproduces the banked Chern–Simons value $CS = \pm 0.07703818$, and this value is a mirror pair — its two signs are the two orientations — governed by the B289 sign law. The non-geometric vacua are likewise class data. No vacuum encodes which knot was filled.*

The pattern is uniform across the invariant battery: geometry, trace field, Chern–Simons, quantum invariants — each is a function of the child’s class, and each is blind to the parent. This is not a failure of resolution; it is the theorem. Surgery is a quotient, and the quotient map has a large kernel: everything the parents differed by, save at most one bit, lies in it.

4 The census and the orientation signature (non-uniform)

Theorem 4.1 (the $(5,1)$. $4_1(5,1)$ and $5_2(5,1)$ are the same underlying manifold $m003(-2,3)$, but as oriented manifolds they are mirror images. The evidence is arithmetic and geometric at once: both have trace field $\mathbb{Q}[x]/(x^4 - x - 1)$, but with **complex conjugate** geometric embeddings (root $-0.2481 + 1.0340i$ for $4_1(5,1)$, root $-0.2481 - 1.0340i$ for $5_2(5,1)$), and their Chern–Simons values are the mirror pair ± 0.07703818 . Here the orientation bit is the one datum that distinguishes the two fillings.

Theorem 4.2 (the triple is orientation-preserving — the law is not uniform). Across the in-window census of surgery collisions on twist knots, a triple also occurs: $4_1(1,2) = 5_2(-1,1) = 6_1(1,1)$, a $\mathbb{Z}$ – homology sphere ($H_1 = 0$, volume 1.3985089). These three fillings are the same oriented manifold — equal Chern–Simons ($CS = -0.246607$, not a $\pm$ pair) — so the collision is orientation-**preserving**. Here orientation distinguishes nothing: three distinct parents, one oriented child, and not even a mirror between them.

The census is the empirical spine of the paper, and it refutes the clean law we first proposed. It is *not* true that every collision on this family is a mirror pair. What is true is weaker and still sharp: whenever two (or more) knots of the family collide to a common child, the parents agree on *everything measurable about the child*, and the only datum on which they can possibly disagree is orientation — a single $\mathbb{Z}/2$ — which they sometimes do (the $(5,1)$ mirror pair) and sometimes do not (the orientation-preserving triple).

5 The inversion law

The inversion law (corrected). Surgery on the figure-eight family forgets the parent knot’s arithmetic entirely — trace field, Alexander polynomial, genus, chirality-as-knot — and the residual distinction between colliding parents is *at most* a single $\mathbb{Z}/2$ datum, the orientation of the child. That bit survives non-uniformly: some collisions are mirror pairs (orientation the sole surviving distinction, as at $(5,1)$), and some are orientation-preserving (no surviving distinction at all, as in the twist-knot triple). Equivalently: the map $(\text{knot}, \text{slope}) \mapsto (\text{oriented child})$ has, on collisions, an ambiguity of zero or one bit — never more — and the bit, when present, is the mirror.

The law is stated for what it forbids as much as for what it keeps. It forbids reading the Eisenstein field $\mathbb{Q}(\sqrt{-3})$, the parent’s Alexander polynomial, the parent’s genus, or the parent’s chirality-as-knot out of the child. What it may keep is one bit, and no more; and that bit is the same $\mathbb{Z}/2$ the whole program keeps under its other operations — orientation, the residue, the breath (Paper 3) — here in its surgical clothing, the mirror in a Dehn filling. The honest headline is not “the object cannot be made to forget which way it turns.” It is: *surgery forgets the parent almost completely, and whether it remembers even the handedness depends on the collision.*

6 What is classical, and what remains open

Classical: cosmetic-surgery and common-filling phenomena (Brakes; Livingston; the twist-knot surgery literature); the class-invariance of trace fields under commensurability (Neumann–Reid); the B289 Chern–Simons sign law’s geometric origin (the deck action); the conjugate-embedding diagnostic for mirror images (Maclachlan–Reid). Ours (scoped, pending the gate): the systematic laundering of the figure-eight’s Eisenstein field from its child (with the child’s quartic trace field having no quadratic subfield); the *non-uniform* orientation residue across the census (the (5,1) mirror pair versus the orientation-preserving triple); the triple collision as a $\mathbb{Z}$ –*homology* sphere. Open: (i) a *characterization* of which collisions are orientation-reversing and which are orientation-preserving — the (p,1) versus multi-slope pattern seen here is not yet a theorem; (ii) whether the laundering extends beyond twist knots; (iii) an intrinsic proof — from the surgery formula rather than the census — that the ambiguity is at most the mirror.

7 Appendix: reproducibility

Census filter and collision enumeration B467_family_residue_wall/census.py, f3_wall.py; the child’s vacua and Chern–Simons B434/B458 engines; the B289 sign law. Revision verification (2026-07-08): trace fields of $4_1, 5_2, 4_1(5,1), 5_2(5,1)$ via *SnapPy/Sage* find_field — $x^2-x+1, x^3-x-1, x^4-x-1, x^4-x-1$ respectively; the conjugate geometric embeddings $-0.2481 \pm 1.0340i$ confirming the (5,1) mirror pair; the triple’s equal volume 1.3985089 and $H_1 = 0$ confirming a common oriented $\mathbb{Z}$ –*homology* sphere. The census solution-type filter correction (an earlier over-strict filter excluded genuine hyperbolic fillings, including the $p = 5$ child) is recorded in the corrections ledger, as is the $v1 \rightarrow v2$ correction (the $\mathbb{Q}(\sqrt{5}) \rightarrow \mathbb{Q}(\sqrt{-3})$ field fix and the orientation-law weakening).

References

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